# Subsystem — External Watchdog

**Role:** External watchdog module — monitors ADCS health, resets RP2040 via RUN pin on timeout
**Interface:** I2C0 (GPIO4/5) + CLK output (GPIO19)
**I2C address:** 0x55 (watchdog side — confirm before deployment)
**Files:** `include/watchdog.h` · `src/watchdog.cpp`

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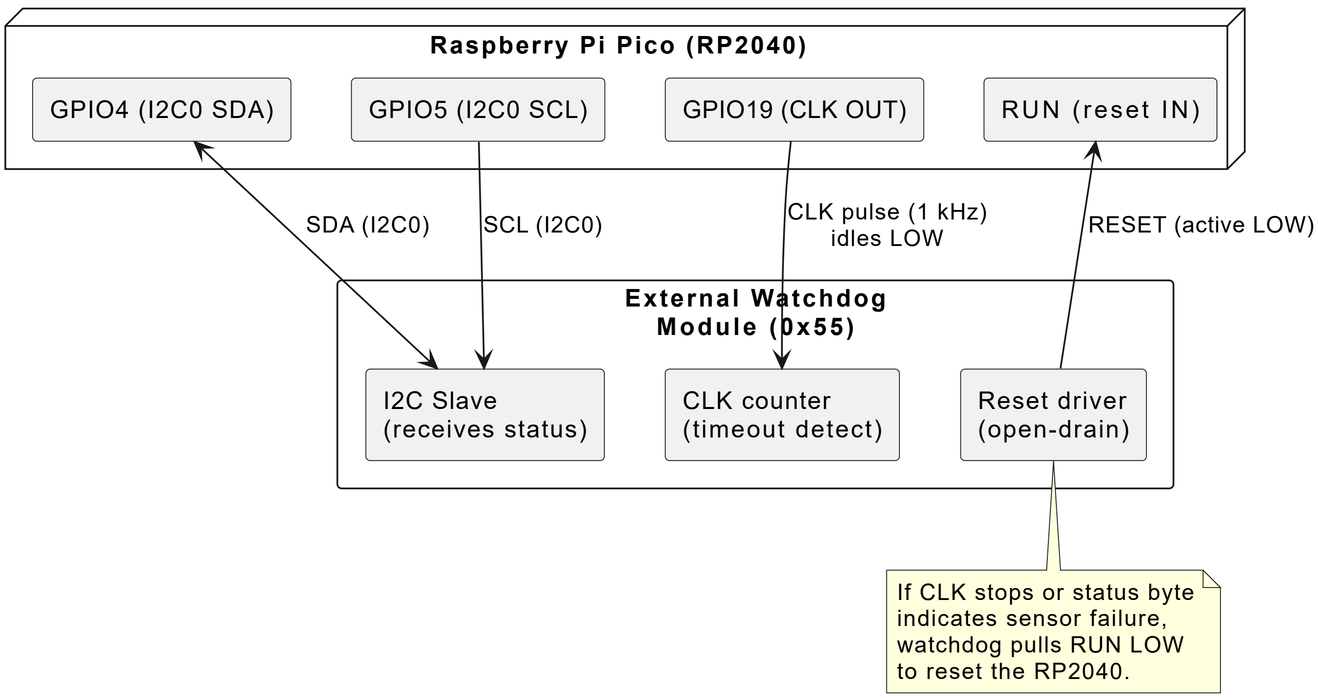
## 1. Hardware Connections




| GPIO | Role | Direction | State |
|---|---|---|---|
| GPIO4 | I2C0 SDA | bidirectional | managed by `Wire` (default Mbed Pico pins) |
| GPIO5 | I2C0 SCL | output | managed by `Wire` |
| GPIO19 | CLK output | OUTPUT | idles LOW; pulsed once per `read()` |
| RUN | hardware reset | INPUT | pulled LOW by watchdog on timeout |

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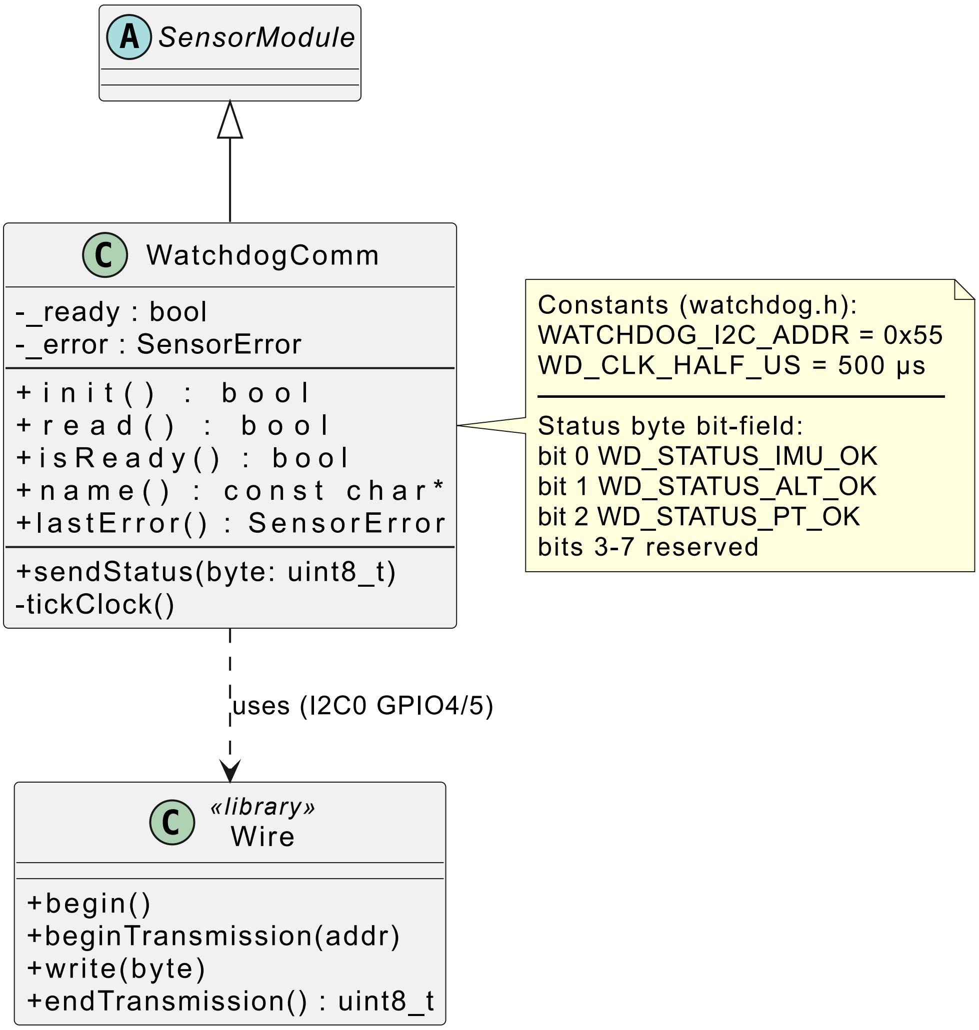
## 2. Class Diagram

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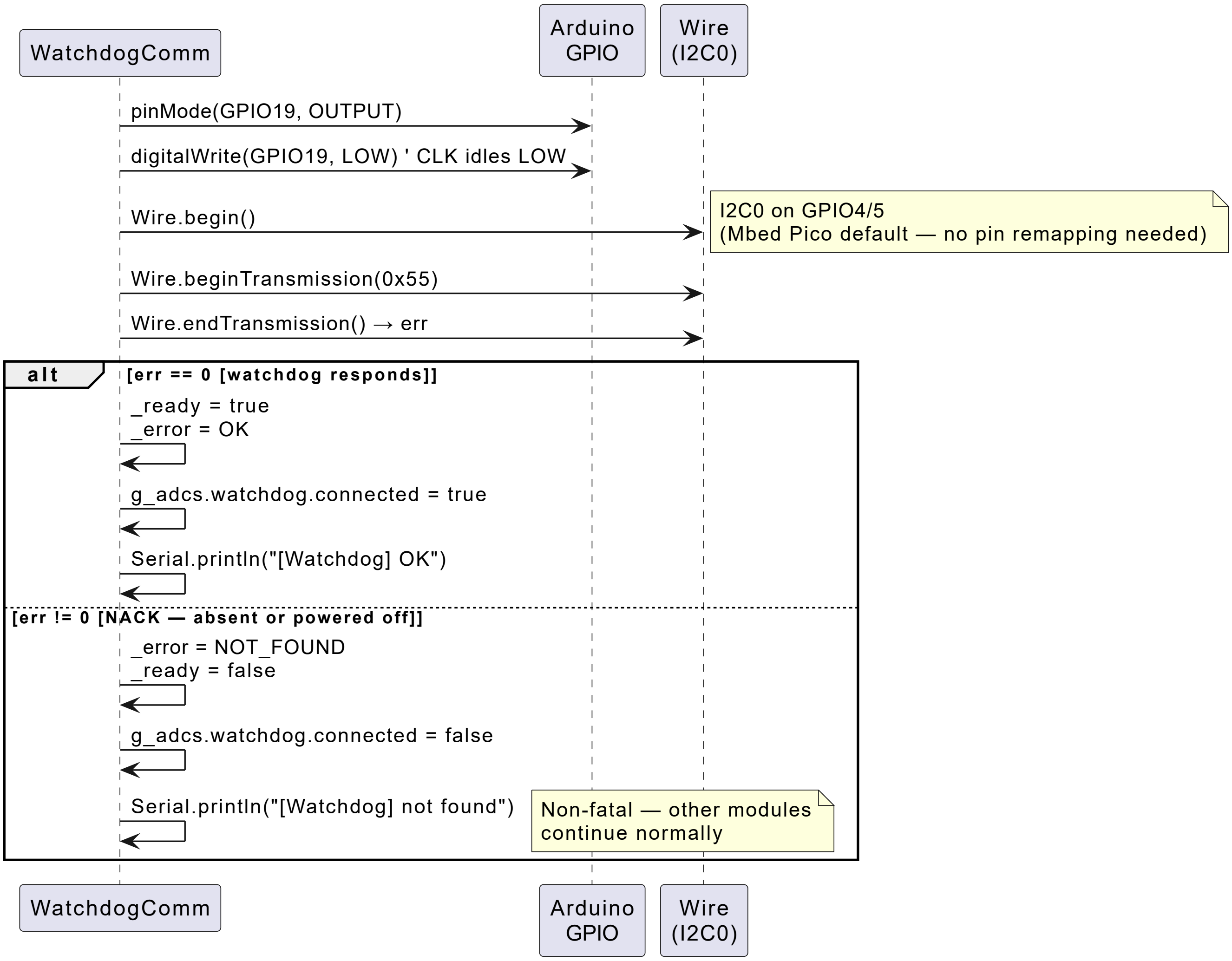
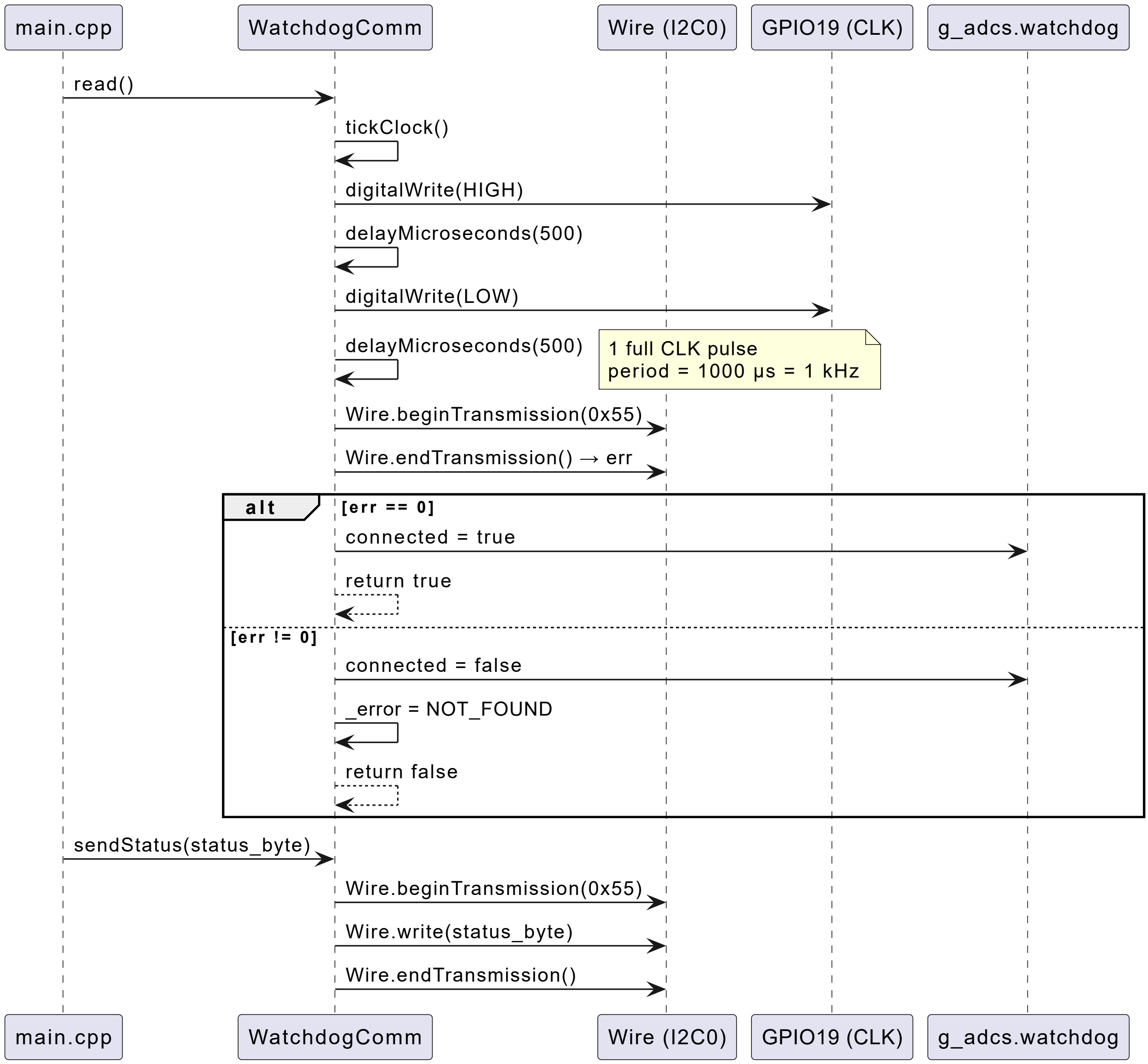
## 3. Initialization Sequence — `init()`




## 4. Per-cycle Sequence — `read()` + `sendStatus()`

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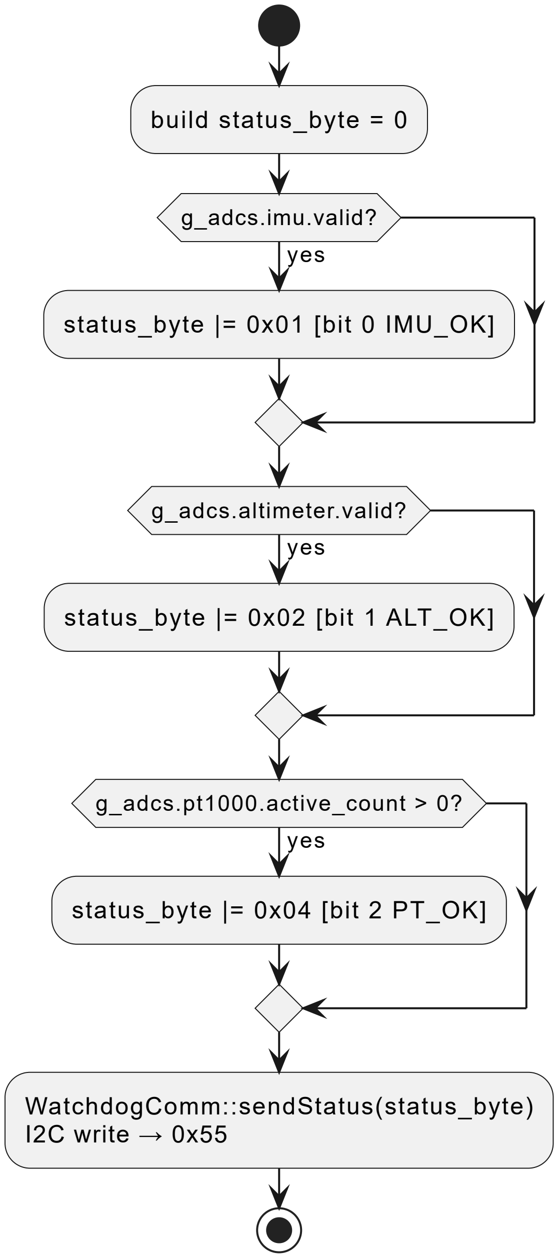
## 5. Status Byte — `sendStatus()`




| Bit | Mask | Meaning |
|---|---|---|
| 0 | `0x01` (`WD_STATUS_IMU_OK`) | `g_adcs.imu.valid == true` |
| 1 | `0x02` (`WD_STATUS_ALT_OK`) | `g_adcs.altimeter.valid == true` |
| 2 | `0x04` (`WD_STATUS_PT_OK`) | `g_adcs.pt1000.active_count > 0` |
| 3–7 | — | reserved |

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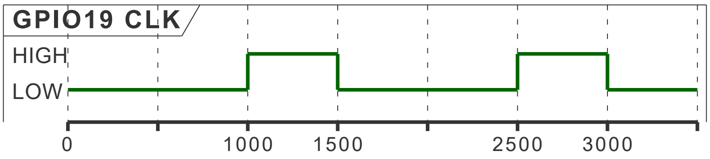
# 6. CLK Signal Timing

**GPIO19 CLK — WD_CLK_HALF_US = 500 µs · period = 1 ms · one pulse per loop() cycle (1 s)**




The CLK pin idles LOW. One pulse is sent at the start of each `read()` call. The watchdog uses the absence of this pulse (over its timeout period) to trigger a reset.